SITING XU

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Research Interest

EDUCATION

Duke University, Durham, NC Sep 2025 – May 2027
M.S. in Electrical and Computer Engineering
***GPA**:4.00* | **Relevant Courses**: *Math for Machine Learning, Practical Machine Learning, Programming and Data Structure for Machine learning*

Shanghai Jiao Tong University, Shanghai, China Sep 2021 - Jun 2025
B.S. in Mathematics and Applied Mathematics & B.Eng. Naval Architecture and Ocean Engineering (dual degree)
***GPA**:3.86* | **Relevant Courses**: *Artificial Intelligence, Big Data Analysis, Stochastic Processes, Numerical Solution of Differential Equations, Theoretical Mechanics, Mechanics of Materials, Principles and Control of Hybrid Power Systems, Fundamentals of Intelligent Ships-I, Innovative Practices in Intelligent Ships*

PUBLICATIONS

Shuai Huang, Tie Li, Ye Wang, Wei He, **Siting Xu,** Hang Cao, Shiyan Li, Run Chen, Ping Yi, Xinran Wang. *A Novel Digital Twin Framework with Hardware-in-the-Loop for Engine Systems*. Accepted for publication in **IEEE Transactions on Intelligent Transportation Systems**

Siting Xu, Shuai Huang, Cheng Meng, et al. *Digital Twin-Oriented Ammonia-Diesel Dual-Fuel Combustion Model and Automatic Calibration Method.* Accepted for the 11th International Conference on Modeling and Diagnostics for Advanced Engine Systems (**COMODIA 2025**)

EXPERIENCE

Differentiable Simulation Framework for Binary Material Optimization in Soft Robotic Fish Sep 2025 - Jul 2026
Role： Graduate Researcher ; General Robotics Lab, Duke University, Durham, NC
Mentor: Boyuan Chen, David Cho
Keywords: soft robotics, differentiable physics, binary material optimization, implicit & semi-implicit Solvers
Developed differentiable simulation tools for soft robotic fish research, focusing on binary material optimization, residual physics learning, and hydrodynamic modeling.

- Extended the DiffPD C++ core to support heterogeneous element-wise materials, including elastic energy, force, and force differential computations.
- **Developed a binary material optimization pipeline** using surrogate gradients for discrete soft/stiff assignments, and evaluated material layouts for maximizing the tip oscillation amplitude of actuated beams.
- Compared DiffPD and Newton solvers through cantilever-beam experiments, assessing runtime, deformation response, and sensitivity to time-step size and material stiffness.
- **Refined residual physics learning** through curriculum learning and loss-function modifications, reducing trajectory error from 0.233 mm to 0.041 mm (82%) in a representative simulation-to-simulation test, and evaluating performance against experimental trajectories.
- **Implemented Lighthill-inspired added-mass and resistive-force models**, analyzed tail frequency responses, and calibrated empirical drag/thrust coefficients against a physics-based simulation baseline at five test frequencies spanning 2–18 Hz.

Foot Placement for Humanoid Locomotion — Unitree G1 Aug 2026 – Present
Role: Independent Research Project
Keywords: reinforcement learning, PPO, humanoid locomotion, Isaac Sim
Developed a reinforcement learning framework for Unitree G1 humanoid locomotion using ASAP/HumanoidVerse and Isaac Sim, focusing on precise foot placement and alternating stepping for straight walking and turning. Implemented a PPO-based training pipeline with curriculum learning and customized reward functions.

- Designed a parallel reinforcement learning environment for Unitree G1 with 23 actuated DoFs and 512 parallel environments, enabling swing-foot tracking of commanded 3D positions and yaw angles while maintaining stance-foot stability.
- **Developed a gait state machine** with an initial standing phase, target-based foot switching, and timeout handling to coordinate alternating steps and prevent failed steps from disrupting planned turning trajectories.
- **Implemented curriculum learning and configurable foot-placement target generators**, including randomized step length, width, and heading commands, as well as fixed-step circular trajectories for continuous turning.
- **Developed customized reward functions** to encourage accurate foot placement, stable upper-body posture, and controlled roll/pitch motion while allowing unrestricted yaw rotation for turning.

Digital Twin Client Development for Marine Power Systems Feb 2023 - Oct 2024

Role：Research Assistant ; Laboratory of Marine Power Plant and Automation at Shanghai Jiao Tong University, Shanghai, China

Mentor: Shuai Huang, Tie Li

Keywords: AI modeling, real-time systems, NSGA-II optimization, sensor fusion, QT development

Developed a digital twin client for a single-cylinder engine with fully measurable state parameters to monitor engine performance and fault, achieving an average error of 3.92% in peak pressure prediction. The project involved integrating hardware and software, constructing digital twin hardware systems, developing a virtual client in QT, creating detailed geometric and physical models, and building a fault sample information library for intelligent fault diagnosis.

- Constructed **a digital twin hardware system, integrating sensors with an industrial control computer to enable accurate, real-time data collection.**
- Developed a virtual client in the QT environment, featuring functionalities for real-time data monitoring and control, including start, work, and exit operations.
- Built computational model to simulate thermodynamic state parameters, displaying real-time in-cylinder pressure curves in the virtual client.
- Established a port communication receiver program for robust data flow monitoring, ensuring seamless transmission of sensor data to the industrial control computer.
- **Developed an automatic calibration method based on NSGA-II** to optimize 33 parameters of the model, achieving an average error of 3.92% in peak pressure compared to measured results.
- Addressed challenges in QT programming, port communication, and model integration by defining transmission protocols, optimizing model coefficients, and ensuring compatibility between hardware and software.

Institute of Automation, Chinese Academy of Sciences Jul 2024 - Sep 2024

Role：Intern Researcher; Computer Vision for Safety Monitoring in Autonomous Systems

- Designed and implemented **a YOLOv5-based detection model to identify unsafe behaviors of miners**, such as not wearing safety helmets, in complex underground environments.
- **Implemented a CLAHE-based preprocessing pipeline, increasing detection accuracy to 92% on low-light, high-noise image datasets.**
- Conducted model training and evaluation on small and medium datasets, comparing the performance of YOLOv5 variants and optimizing parameters to enhance detection efficiency.
- Proposed improvements to address underfitting and increase robustness in real-world scenarios, including data augmentation and architectural adjustments.

AWARDS

- *ABS (American Bureau of Shipping) Scholarship, December 2023*
- *Meritorious Winner, Mathematical Contest in Modelling (MCM), United States, May 2023*
- *Second Prize, Shanghai Municipal Competition (Undergraduate Group), National College Student Mathematical Contest in Modelling, November 2022*

SKILLS

Programming: IsaacSim, C++, Python, MATLAB and R